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Algebra that does the work

Level 2 Intermediate · Algebra

Factorising, the difference of two squares, substitution and combining equations, so that the structure of an expression does the arithmetic for you.

By the end you can: You will be able to spot a common factor or a difference of two squares before expanding anything; You will be able to handle symmetric problems by working with the sum and product instead of the unknowns; You will be able to solve contest systems and integer equations by adding, multiplying or factorising.

Lesson

Try this before reading on: evaluate

$$\frac{2026^3 - 2026}{2025 \times 2027}.$$

You could multiply everything out. The cube of 2026 has ten digits, and after subtracting you would still face a long division. Instead, look at the numerator. Both terms contain a factor of 2026, so $2026^3 - 2026 = 2026\left(2026^2 - 1\right)$. The bracket is a square minus one, which factorises as $(2026 - 1)(2026 + 1) = 2025 \times 2027$. That is the denominator, it cancels, and the answer is 2026. No arithmetic at all.

That is the spirit of this unit. Contest algebra usually has a hidden structure, and finding it is quicker and safer than grinding. We will collect four tools: factorising, the difference of two squares, substitution, and combining equations before solving them.

Factorise, don't expand

Expanding is the mechanical direction. Factorising is the direction that tells you things: a product shows you divisibility, shows you when an expression is zero, and shows you what cancels. These identities should be automatic.

Core identities. For all numbers a, b, x, y:

$$(a+b)^2 = a^2 + 2ab + b^2, \qquad (a-b)^2 = a^2 - 2ab + b^2, \qquad a^2 - b^2 = (a-b)(a+b),$$

$$xy + x + y + 1 = (x+1)(y+1).$$

Each one is checked by expanding the right-hand side. The last looks innocent, but it is the key to a whole family of integer equations. If you see xy, x and y together, try to complete a product by adding a constant. For instance $(x-2)(y-3) = xy - 3x - 2y + 6$, so $xy - 3x - 2y$ is the same as $(x-2)(y-3) - 6$.

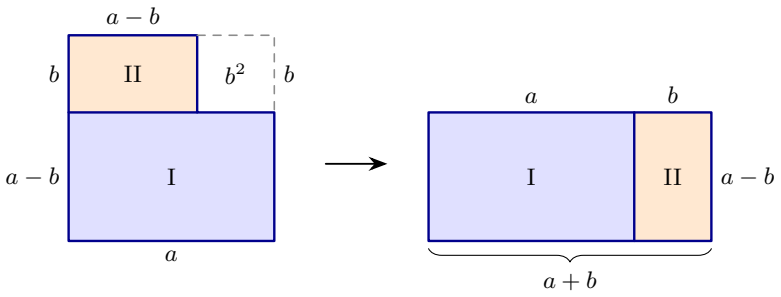
The habit to build is simple: before expanding anything, ask whether there is a common factor, a perfect square or a difference of squares hiding in it.

The difference of two squares

The identity $a^2 - b^2 = (a-b)(a+b)$ deserves its own section, because it turns up constantly, with numbers as well as letters. The picture shows why it is true. Remove a $b \times b$ square from the corner

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of an $a \times a$ square, cut the L-shape that is left into two rectangles, and slide one of them round to make a single rectangle of width $a+b$ and height $a-b$.



It gets used in three ways.

- Arithmetic: $97 \times 103 = 100^2 - 3^2 = 9991$, and $2025^2 - 2024^2 = 2025 + 2024 = 4049$.
- Square roots: $(\sqrt{m} - \sqrt{n})(\sqrt{m} + \sqrt{n}) = m - n$, so knowing $\sqrt{m} - \sqrt{n}$ tells you $\sqrt{m} + \sqrt{n}$ as well.
- Integers: to solve $a^2 - b^2 = N$ in integers, factorise N.

The third use gives a neat theorem.

Which integers are differences of two squares? An integer N can be written as $a^2 - b^2$ with a and b integers if and only if N is odd or a multiple of 4.

Proof. The numbers $a-b$ and $a+b$ differ by $2b$, which is even, so they are both odd or both even. If both are odd, then $N = (a-b)(a+b)$ is odd. If both are even, then N is a multiple of $2 \times 2 = 4$. So a number that is 2 more than a multiple of 4, such as 6 or 10, is never a difference of two squares. Conversely, every odd number $2k+1$ equals $(k+1)^2 - k^2$, and every multiple of four $4k$ equals $(k+1)^2 - (k-1)^2$. □

Substitution: name the lump

If the same expression keeps appearing, give it a name. The equation $\left(x^2+x\right)^2 - 8\left(x^2+x\right) + 12 = 0$ looks like a quartic, but with $t = x^2 + x$ it reads $t^2 - 8t + 12 = 0$, so $t = 2$ or $t = 6$. Then $x^2 + x = 2$ gives $x = 1$ or $x = -2$, and $x^2 + x = 6$ gives $x = 2$ or $x = -3$. A frightening quartic became two easy quadratics.

The most important case of this idea is symmetry. When a problem treats x and y in the same way, work with the sum $s = x + y$ and the product $p = xy$ rather than with x and y themselves.

Symmetric expressions. If $s = x + y$ and $p = xy$, then

$$x^2 + y^2 = s^2 - 2p, \qquad x^3 + y^3 = s^3 - 3ps, \qquad \frac{1}{x} + \frac{1}{y} = \frac{s}{p} \quad (p \neq 0).$$

Conversely, given s and p, the numbers x and y are the two roots of $t^2 - st + p = 0$, so real x and y exist exactly when $s^2 \geq 4p$.

The first identity is $(x+y)^2$ expanded. For the second, $(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3 = x^3 + y^3 + 3xy(x+y)$. For the third, put the fractions over the common denominator xy. For the last

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statement, $(t-x)(t-y) = t^2 - st + p$, and a quadratic has real roots exactly when its discriminant, here $s^2 - 4p$, is not negative.

The point is that you often never need x and y. If $x + y = 4$ and $xy = 1$, then $x^2 + y^2 = 16 - 2 = 14$ at once, whereas finding $x = 2 \pm \sqrt{3}$ and squaring is slower and invites slips.

Simultaneous equations: combine before you substitute

Substituting one equation into another always works in principle, but contest systems are designed to reward looking at the whole system first. Three moves cover most of them.

- **Add everything.** If each unknown appears equally often, adding gives the total. From $a + b = 9$, $b + c = 13$ and $c + a = 14$ we get $2(a + b + c) = 36$, so $a + b + c = 18$, and then $c = 18 - 9 = 9$, $a = 5$ and $b = 4$.
- **Subtract to cancel.** Subtracting two similar equations often leaves something that factorises.
- **Multiply everything.** When the equations are products, multiplying them together plays the role that adding plays for sums.

Also read what the question actually asks for. It is often a combination such as $x + y$ or xy, which can be found without solving for each unknown.

Example 1. Find all pairs of positive integers (a, b) with $a^2 - b^2 = 77$.

Solution.
The first instinct is to list squares and look for two that differ by 77. Since $81 - 4 = 77$, the pair $(9, 2)$ works. But when do you stop searching, and how do you know nothing was missed? Factorising answers both questions.
Write the equation as $(a - b)(a + b) = 77$. As a and b are positive, $a + b$ is positive, so $a - b$ is positive as well, and $a - b$ is smaller than $a + b$. The factor pairs of 77 are 1×77 and 7×11. From $a - b = 1$ and $a + b = 77$ we get $a = 39$, $b = 38$. From $a - b = 7$ and $a + b = 11$ we get $a = 9$, $b = 2$.
So there are exactly two pairs, $(39, 38)$ and $(9, 2)$. The factorisation turned an open-ended search into a short list of cases. Had N been even, we would also have needed $a - b$ and $a + b$ to have the same parity, which rules out some factor pairs.

Example 2. The real number x satisfies $x + \dfrac{1}{x} = 3$. Find $x^2 + \dfrac{1}{x^2}$ and $x^3 + \dfrac{1}{x^3}$.

Solution.
You could solve for x: multiplying by x gives $x^2 - 3x + 1 = 0$, so $x = \frac{3 \pm \sqrt{5}}{2}$. Now cube that. It can be done, but it is slow, and one slip ruins it. Instead, notice that x and $\frac{1}{x}$ have sum 3 and product 1. This is the symmetric situation with $s = 3$ and $p = 1$.
Squaring, $\left(x + \frac{1}{x}\right)^2 = x^2 + 2 + \frac{1}{x^2}$, so $x^2 + \frac{1}{x^2} = 9 - 2 = 7$. Cubing,

$$\left(x + \frac{1}{x}\right)^3 = x^3 + 3x + \frac{3}{x} + \frac{1}{x^3} = x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right),$$

so $x^3 + \frac{1}{x^3} = 27 - 9 = 18$.
The cross terms are the whole trick. When you square or cube a number plus its reciprocal, the cross terms are constants or multiples of what you already know. Squaring 7 the same way gives $x^4 + \frac{1}{x^4} = 49 - 2 = 47$, still without ever finding x.

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Example 3. Positive real numbers x, y and z satisfy $xy = 6$, $yz = 15$ and $zx = 10$. Find $x + y + z$.

Solution.
Substitution works: $y = \frac{6}{x}$, so $z = \frac{15}{y} = \frac{5x}{2}$, and then $zx = \frac{5x^2}{2} = 10$ gives $x^2 = 4$. That is fine, but there is a cleaner route which scales to bigger systems.
Each unknown appears in exactly two of the equations, so multiply all three: $(xyz)^2 = 6 \times 15 \times 10 = 900$. Everything is positive, so $xyz = 30$. Now divide by each equation in turn: $z = \frac{xyz}{xy} = \frac{30}{6} = 5$, then $x = \frac{30}{15} = 2$ and $y = \frac{30}{10} = 3$. So $x + y + z = 10$.
This is the multiplicative version of adding all the equations. Notice that positivity was used: without it, $xyz = -30$ would also be possible, giving $x + y + z = -10$.

Example 4. How many pairs of positive integers (x, y) satisfy $xy - 2x - 3y = 6$?

Solution.
A natural first attempt is to make x the subject: $x(y - 2) = 3y + 6$, so $x = \frac{3y+6}{y-2}$. Trying $y = 3, 4, 5, \ldots$ produces some solutions, but again you do not know when to stop. The combination xy, $-2x$, $-3y$ is the signal for the add-a-constant factorisation.
We want $(x-3)(y-2)$, which expands to $xy-2x-3y+6$. So add 6 to both sides: $(x-3)(y-2) = 12$. Now $x - 3$ and $y - 2$ are integers with product 12. Could both be negative? Since $x \geq 1$ and $y \geq 1$, we would have $x - 3 \geq -2$ and $y - 2 \geq -1$, and two negative integers of that size have product at most 2. So both factors are positive, and each positive divisor d of 12 gives exactly one solution, $x = 3 + d$ and $y = 2 + \frac{12}{d}$.
As 12 has the six positive divisors $1, 2, 3, 4, 6, 12$, there are 6 pairs: $(4, 14)$, $(5, 8)$, $(6, 6)$, $(7, 5)$, $(9, 4)$ and $(15, 3)$. The first attempt was not wrong, by the way: $\frac{3y+6}{y-2} = 3 + \frac{12}{y-2}$ says the same thing. Either way, it is the factorisation that makes the count complete.

Common traps.

- $(a + b)^2$ is not $a^2 + b^2$. The cross term $2ab$ is usually the whole point of the question.
- Cancelling terms instead of factors. $\frac{a^2+b^2}{a+b}$ does not simplify: only a factor of the whole top and the whole bottom can cancel.
- Dividing by something that might be zero. From $x^2 = 5x$ you cannot conclude $x = 5$. Factorise to $x(x - 5) = 0$ and keep both roots.
- In integer factor problems, forgetting the negative factor pairs, or forgetting that $a - b$ and $a + b$ must have the same parity.
- Finding s and p but never checking that real x and y exist, which needs $s^2 \geq 4p$.

How to spot this in a contest

Large numbers sitting next to each other (like 2025, 2026, 2027) or close to a round number are nearly always a factorisation in disguise: call the middle one n and write the others as $n \pm 1$. A question that gives you $x + y$ and xy and asks for $x^2 + y^2$, $x^3 + y^3$ or $\frac{1}{x} + \frac{1}{y}$ wants you to stay with the sum and product. An equation in positive integers that mixes xy with x and y wants the add-a-constant factorisation. Three or more equations with a cyclic pattern want to be added or multiplied. A repeated lump wants a name.

Above all, whenever you are about to expand something large, stop for ten seconds and look for a common factor or a difference of squares first. On the IMC and the AMC 10 that pause is often the difference between a two-minute question and a ten-minute one.

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Practice

Try each problem before turning to the hints and solutions at the back.

1. ⋆ Evaluate $\dfrac{2024^2 - 576}{2000}$.

2. ⋆⋆ Find the sum of the squares of all the real solutions of

$$\left(x^2 - 5x\right)^2 - 2\left(x^2 - 5x\right) - 24 = 0.$$

3. ⋆⋆ Real numbers a and b satisfy $a + b = 6$ and $ab = 4$. Find $a^3 + b^3$.

4. ⋆⋆⋆ How many of the integers $1, 2, 3, \ldots, 200$ can be written in the form $a^2 - b^2$, where a and b are both *positive* integers?

5. ⋆⋆⋆ Positive integers x and y satisfy $xy = 2x + 5y + 10$. Find the sum of all possible values of x.

6. ⋆⋆⋆⋆ Real numbers x and y satisfy

$$x + y + xy = 11 \qquad \text{and} \qquad x^2y + xy^2 = 30.$$

What is the largest possible value of $x^2 + y^2$?

Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

1. AMC 10A 2016 Problem 1 ⋆
2. UKMT IMC 2018 Question 10 ⋆
3. UKMT IMC 2017 Question 12 ⋆⋆
4. AMC 10A 2017 Problem 5 ⋆⋆
5. AMC 10B 2019 Problem 6 ⋆⋆
6. AMC 10A 2018 Problem 10 ⋆⋆
7. AMC 10A 2020 Problem 14 ⋆⋆⋆
8. UKMT IMC 2017 Question 16 ⋆⋆⋆
9. AMC 10A 2021 Problem 9 ⋆⋆⋆
10. AMC 10A 2021 Problem 10 ⋆⋆⋆
11. AMC 10A 2019 Problem 19 ⋆⋆⋆⋆
12. AMC 10A 2020 Problem 21 ⋆⋆⋆⋆

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Hints

1. Is 576 a perfect square?
2. Give the repeated expression a name and solve the quadratic you get first.
3. Expand $(a+b)^3$ and see which part of it you already know.
4. Write $N=(a-b)(a+b)$. What does $b\geq 1$ add to the conditions on the two factors?
5. Move everything to one side and add a constant so that the left-hand side becomes a product of two brackets.
6. Both equations are symmetric. Rewrite them in terms of $s=x+y$ and $p=xy$, and remember to check that real x and y exist.

Solutions

Problem 1

Answer: 2048

Since $576=24^2$, the numerator is a difference of two squares: $2024^2-24^2=(2024-24)(2024+24)=2000\times 2048$. Dividing by 2000 leaves 2048.

Problem 2

Answer: 54

Let $t=x^2-5x$. The equation becomes $t^2-2t-24=0$, which factorises as $(t-6)(t+4)=0$, so $t=6$ or $t=-4$. If $t=6$, then $x^2-5x-6=0$, that is $(x-6)(x+1)=0$, giving $x=6$ or $x=-1$. If $t=-4$, then $x^2-5x+4=0$, that is $(x-1)(x-4)=0$, giving $x=1$ or $x=4$. These four real solutions are distinct, and the sum of their squares is $36+1+1+16=54$.

Problem 3

Answer: 144

Expanding, $(a+b)^3=a^3+b^3+3ab(a+b)$. So $a^3+b^3=(a+b)^3-3ab(a+b)=216-3\times 4\times 6=216-72=144$. Such real numbers do exist, because $(a+b)^2=36$ is at least $4ab=16$; they are $3\pm\sqrt{5}$.

Problem 4

Answer: 148

Write $N=uv$ with $u=a-b$ and $v=a+b$. Then u and v are positive with the same parity, and $b\geq 1$ means $v-u=2b\geq 2$, so u is smaller than v. Conversely, if $N=uv$ with u smaller than v and both of the same parity, then $a=\frac{u+v}{2}$ and $b=\frac{v-u}{2}$ are positive integers with $a^2-b^2=N$.

If N is odd and at least 3, the factorisation $1\times N$ works, so every odd number from 3 to 199 is allowed: that is 99 numbers. The number 1 fails, because its only factorisation is 1×1. If N is even, both factors must be even, so N is a multiple of 4, say $N=4k$. The factorisation $2\times 2k$ works when $k\geq 2$, while $N=4$ fails because its only factorisation into two even numbers is 2×2. So the multiples of 4 from 8 to 200 are allowed: that is 49 numbers. Numbers that are 2 more than a multiple of 4 never work. The total is $99+49=148$.

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Problem 5

Answer: 72

Rearranging gives $xy - 2x - 5y = 10$. Since $(x-5)(y-2) = xy - 2x - 5y + 10$, adding 10 to both sides gives $(x-5)(y-2) = 20$. As x and y are positive, $x - 5 \geq -4$ and $y - 2 \geq -1$. If both factors were negative, we would need $y - 2 = -1$ and so $x - 5 = -20$, which is impossible. So both factors are positive, and $x - 5$ is a positive divisor of 20, namely $1, 2, 4, 5, 10$ or 20. Each gives a valid y, and the possible values of x are $6, 7, 9, 10, 15, 25$, whose sum is 72.

Problem 6

Answer: 26

Let $s = x + y$ and $p = xy$. The equations become $s + p = 11$ and $ps = 30$, since $x^2y + xy^2 = xy(x+y)$. So s and p are the two roots of $t^2 - 11t + 30 = 0$, which factorises as $(t-5)(t-6) = 0$. Hence either $s = 5$ and $p = 6$, or $s = 6$ and $p = 5$.

Real x and y exist when $s^2 \geq 4p$. In the first case $25 \geq 24$, and in the second $36 \geq 20$, so both cases occur (in fact they give $\{x, y\} = \{2, 3\}$ and $\{1, 5\}$). Since $x^2 + y^2 = s^2 - 2p$, the two cases give $25 - 12 = 13$ and $36 - 10 = 26$. The largest possible value is 26.

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